

BIOS 755: Covariance Pattern Analysis and the General Linear Model

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Treatment of Lead-Exposed Children (TLC) Trial

- ▶ The methods we'll discuss today will add a covariance matrix to standard linear regression.
- ▶ For a covariance matrix to be meaningful, it is easiest if the data have well-defined time-points.
- ▶ As a result, we'll reflect on the TLC example.
- ▶ Part of the reason we'll do this is because the methods we will discuss are most useful for data that are (at least) planned to have **balanced** data.
- ▶ For observational data, for example, the HIV study, covariance pattern analysis is not recommended (linear mixed models are).

Treatment of Lead-Exposed Children (TLC) Trial

- ▶ Facts of the trial:
 - ▶ Randomized trial,
 - ▶ 100 children randomized to placebo or Succimer,
 - ▶ measures of blood lead level at baseline, 1, 4, and 6 weeks.
- ▶ For randomized trials, there is no need to adjust for confounders (e.g., age, race/ethnicity, SES).

General Linear Model

- ▶ For each observation, Y_{ij} , assume we have an associated set of covariates

$$\mathbf{X}_{ij} = \{1, X_{ij1}, X_{ij2}, \dots, X_{ijp}\}$$

- ▶ Information about the time of the observations, treatment group, age, biomarkers, and other predictor variables can be expressed through a vector of covariates.
- ▶ The one represents the intercept.
- ▶ How we structure $\mathbf{X}$ will be discussed in later lectures.

General Linear Model

- ▶ The general linear model can be written as

$$Y_{i1} = \beta_0 + \beta_1 X_{i11} + \beta_2 X_{i12} + \dots + \beta_p X_{i1p} + e_{i1}$$

$$Y_{i2} = \beta_0 + \beta_1 X_{i21} + \beta_2 X_{i22} + \dots + \beta_p X_{i2p} + e_{i2}$$

$$\vdots = \vdots$$

$$Y_{in} = \beta_0 + \beta_1 X_{in1} + \beta_2 X_{in2} + \dots + \beta_p X_{inp} + e_{in}$$

- ▶ We can summarize this to

$$Y_{ij} = \beta_0 + \sum_{k=1}^p X_{ijk} \beta_k + e_{ij} \quad \text{for } j = 1, 2, \dots, n$$

General Linear Model

- ▶ When we remove the error term e_{ij} we get the predicted values

$$E(Y_{ij}) = \hat{Y}_{ij} = \mu_{ij} = \beta_0 + \sum_{j=1}^p X_{ij}\beta_j$$

- ▶ The difference between the predicted and observed values are the residuals or error

$$Y_{ij} - \hat{Y}_{ij} = e_{ij}$$

General Linear Model

- ▶ With longitudinal data, we expect the error terms, e_{ij} , to be correlated within individuals.
- ▶ For example, if an individual has a large positive error term in the first observation, i.e.,

$$Y_{i1} - \hat{Y}_{ij} = e_{i1} > 0 \quad \text{is large}$$

then, what would you expect the error term of the second observation to be?

- ▶ So, in longitudinal data, we want to allow for

$$\text{corr}(e_{ij}, e_{ik}) \neq 0$$

for all j and k .

Covariance Matix

- This leads to a covariance matrix for $\mathbf{e}_i$

$$\text{Cov}(\mathbf{e}_i) = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \dots & \sigma_{1n} \\ \sigma_{21} & \sigma_2^2 & \dots & \sigma_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{n1} & \sigma_{n2} & \dots & \sigma_n^2 \end{pmatrix} = \Sigma$$

where $\text{cov}(e_{ij}, e_{ik}) = E(Y_j - \mu_j)(Y_k - \mu_k) = \sigma_{jk}$ with $\sigma_{jj} = \sigma_j^2$.

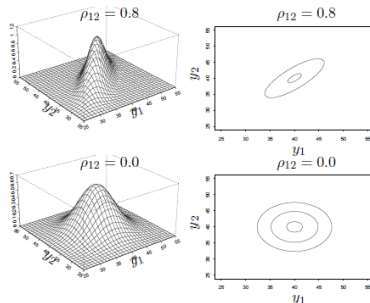
General Linear Model

- ▶ Yet *another* to write the general linear model is

$$\mathbf{Y}_i = \mathbf{X}_i \boldsymbol{\beta} + \mathbf{e}_i$$

where $\mathbf{X}_i = \{\mathbf{X}_{i1}, \dots, \mathbf{X}_{ini}\}$ and $\mathbf{e}_i \sim MVN(\mathbf{0}, \Sigma)$.

- ▶ Recall that Σ is a covariance matrix of the residual error terms.
- ▶ **Main Question:** What Σ to use?



Covariance Structure

When choosing a covariance structure, the important aspects to consider are:

- ▶ Balanced or unbalanced **time points**.
 - ▶ **For unbalanced time points, our options are limited.**
- ▶ Homogeneity or heterogeneity (i.e., is the residual variance equal or not equal over time)?
- ▶ Are there simple forms we can use to represent the correlation?

Covariance Structure

- ▶ The most flexible is the unrestricted or unstructured covariance matrix (**heterogenous** and **no assumptions on correlation**):

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \sigma_{13} & \sigma_{14} \\ \sigma_{21} & \sigma_2^2 & \sigma_{23} & \sigma_{24} \\ \sigma_{31} & \sigma_{32} & \sigma_3^2 & \sigma_{34} \\ \sigma_{41} & \sigma_{42} & \sigma_{43} & \sigma_4^2 \end{pmatrix} \quad \Gamma = \begin{pmatrix} 1 & \rho_{12} & \rho_{13} & \rho_{14} \\ \rho_{21} & 1 & \rho_{23} & \rho_{24} \\ \rho_{31} & \rho_{32} & 1 & \rho_{34} \\ \rho_{41} & \rho_{42} & \rho_{43} & 1 \end{pmatrix}$$

- ▶ The most restrictive covariance matrix is the Independence matrix (**homogeneous** and **no correlation allowed**):

$$\Sigma = \begin{pmatrix} \sigma^2 & 0 & 0 & 0 \\ 0 & \sigma^2 & 0 & 0 \\ 0 & 0 & \sigma^2 & 0 \\ 0 & 0 & 0 & \sigma^2 \end{pmatrix} \quad \Gamma = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Compound Symmetry, Exchangeable

- ▶ Two popular covariance models with this correlation matrix are the:
 - ▶ compound symmetric, and
 - ▶ heterogeneous compound symmetric structure
- ▶ The difference in these structures is whether or not we assume the variance is homogeneous or heterogeneous across time points.

(Homogeneous) Compound Symmetric

- The compound symmetric structure:

$$\Sigma = \begin{pmatrix} \sigma^2 & \rho\sigma^2 & \dots & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 & \dots & \rho\sigma^2 \\ \vdots & \vdots & \ddots & \vdots \\ \rho\sigma^2 & \rho\sigma^2 & \dots & \sigma^2 \end{pmatrix} \quad \Gamma = \begin{pmatrix} 1 & \rho & \rho & \rho \\ \rho & 1 & \rho & \rho \\ \rho & \rho & 1 & \rho \\ \rho & \rho & \rho & 1 \end{pmatrix}$$

also called an exchangeable structure.

Heterogeneous Compound Symmetric

- The heterogeneous compound symmetric structure:

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 & \dots & \rho\sigma_1\sigma_k \\ \rho\sigma_1\sigma_2 & \sigma_2^2 & \dots & \rho\sigma_2\sigma_k \\ \vdots & \vdots & \ddots & \vdots \\ \rho\sigma_1\sigma_k & \rho\sigma_2\sigma_k & \dots & \sigma_k^2 \end{pmatrix} \quad \Gamma = \begin{pmatrix} 1 & \rho & \rho & \rho \\ \rho & 1 & \rho & \rho \\ \rho & \rho & 1 & \rho \\ \rho & \rho & \rho & 1 \end{pmatrix}$$

- What is error? Would it change?

Autoregressive Structure of Order 1 (AR(1))

- ▶ Autoregressive correlation matrix:

$$\Gamma = \begin{pmatrix} 1 & \rho & \rho^2 & \rho^3 & \rho^4 \\ \rho & 1 & \rho & \rho^2 & \rho^3 \\ \rho^2 & \rho & 1 & \rho & \rho^2 \\ \rho^3 & \rho^2 & \rho & 1 & \rho \\ \rho^4 & \rho^3 & \rho^2 & \rho & 1 \end{pmatrix}$$

- ▶ Since ρ is less than one, as we take higher powers of it, the results get closer and closer to zero.
- ▶ As observations get further away (in terms of number of observations) their correlation gets smaller.

Visualize the Correlation Structures

$R =$

	Visit 1	Visit 2	Visit 3	Visit 4
Visit 1	1			
Visit 2	0.7	1		
Visit 3	0.7 ²	0.7	1	
Visit 4	0.7 ³	0.7 ²	0.7	1

Exponential Structure

- ▶ The Exponential Structure uses the time between points in calculating the correlation.
- ▶ The correlation between two points Y_{ij} and Y_{ik} is equal to

$$\rho_{jk} = \exp \left\{ -\frac{|t_{ij} - t_{ik}|}{\theta} \right\}$$

recall that t_{ij} is the time of observation Y_{ij} , similarly for t_{ik} .

- ▶ The parameter θ is estimated, the larger the value of θ the smaller the correlation.

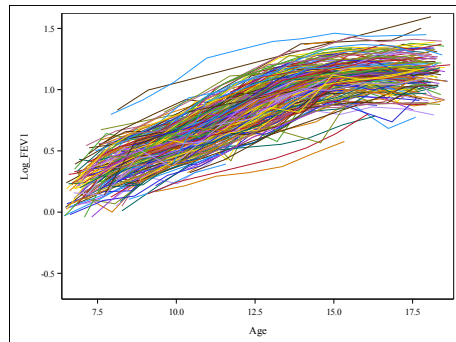
Fitting in SAS

- ▶ SAS can be used to fit many, many covariance structures.
- ▶ Click [here](#) for a full list of covariance matrices.

GO TO EXAMPLE

Balanced or Unbalanced time points

- ▶ One big factor in choosing a covariance matrix is if the time points are balanced or not.
- ▶ In the unstructured correlation matrix ρ_{jk} is the correlation between the j th and k th time points. For example, ρ_{23} measures the correlation between the 2nd and 3rd measurements.
- ▶ For unbalanced data, the 2nd and 3rd measurements could be from ages 8 and 9 for one kid and ages 10 and 16 for another kid.



Unbalanced covariance structures

- Of the covariance matrices we've discussed, only the homogeneous compound symmetric and homogeneous exponential make sense for time points that are unbalanced across subjects.

Unbalanced covariance structures

- ▶ Of the covariance matrices we've discussed, only the homogeneous compound symmetric and homogeneous exponential make sense for time points that are unbalanced across subjects.
- ▶ The exponential can also be used to model the covariate over space.
- ▶ For example, say d_{1k} and d_{2k} are the latitude and longitude of the k th measurement.
- ▶ The correlation between two points Y_j and Y_k is equal to

$$\rho_{jk} = \exp \left\{ -\frac{\sqrt{(d_{1j} - d_{1k})^2 + (d_{2j} - d_{2k})^2}}{\theta} \right\}$$

which is a spatial covariance matrix.

General Linear Model

- ▶ The general linear model can be written as

$$\mathbf{Y}_i = \mathbf{X}_i\boldsymbol{\beta} + \mathbf{e}_i$$

where $\mathbf{X}_i = \{\mathbf{X}_{i1}, \dots, \mathbf{X}_{in_i}\}$ and $\mathbf{e}_i \sim MVN(\mathbf{0}, \Sigma)$.

- ▶ Recall that Σ is a covariance matrix of the residual error terms.
- ▶ When we fit a model to longitudinal data, the main aim is to estimate the unknown parameters, $\boldsymbol{\beta}$ and Σ .
- ▶ The estimates are obtained via maximum likelihood estimation.

Introduction to Maximum Likelihood (ML)

- ▶ The main idea behind Maximum Likelihood (ML) is simple: use as estimates of β and Σ the most probable values (or “likely”) for the observed data.
- ▶ Choose values of β and Σ that maximize the probability of the response variables evaluated at their observed values (or that best predict the observed data).
- ▶ The resulting values, $\hat{\beta}$ and $\hat{\Sigma}$ are called the maximum likelihood estimates (MLE) of β and Σ , and have many appealing properties: consistency, normality, and robustness.

MLE Bias

- MLE of σ^2 has well-known bias in small samples. For ordinary regression with independent errors gives the usual least squares estimator of β , but the ML estimator of σ^2 is

$$E(\hat{\sigma}^2) = \left(\frac{n-k}{n} \right) \sigma^2$$

- As a result, the ML estimate of σ^2 will be biased in small samples and will underestimate σ^2 (on average).

Restricted Maximum Likelihood

- ▶ The theory of restricted (or residual) maximum likelihood (REML) estimation was developed to address this problem.
- ▶ REML uses a penalty term that eliminates the biases in the variance parameter estimates.
- ▶ Estimates of Σ obtained by the REML are preferred in practice. All (I think) softwares will use this as the default.

Restricted Maximum Likelihood

- ▶ **REML should not be used to compare different regression models (i.e., different mean models)** since the penalty term depends upon the regression model specification.
- ▶ Instead, the standard ML log-likelihood should be used for comparing different regression models for the mean.
- ▶ That is why we have to use `method="ML"` when we compare different models for the mean.
- ▶ **The REML can be used to compare different models for the covariance structure.**

Akaike information criterion

- ▶ Akaike information criterion (AIC) judges a model by the value of the negative log-likelihood ($-\log\{L(\beta)\} > 0$).
- ▶ $-\log\{L(\beta)\}$ will increase as more parameters are added to the model so **we penalize by the number of parameters**.
- ▶ AIC is a very general method of evaluating model fit that can be used in a wide variety of statistical procedures.
- ▶ The formula for AIC is

$$-2 \log\{L(\hat{\beta})\} + 2q$$

where q is the number of parameters in the model.

- ▶ We choose the model with the lowest AIC

Bayesian information criterion

- ▶ Bayesian information criterion (BIC) is very similar to AIC
- ▶ BIC penalizes more heavily for adding additional parameters.
- ▶ The formula for BIC is

$$-2 \log\{L(\hat{\beta})\} + q \log(N^*)$$

where N^* is the number of **effective subjects**.

- ▶ We choose the model with the lowest BIC.
- ▶ The penalty for AIC is $2q$ the penalty for BIC is $\log(N^*)q$

$$2 < \log(N^*) \quad N^* \geq 8$$

Likelihood Ratio Tests

- ▶ Two models are nested (bigger v.s smaller)
- ▶ Under the null hypothesis, the difference of $-2 \log\{L(\hat{\beta})\}$ between the full and reduced model follows a χ^2 distribution with $df = \Delta df$
- ▶ A likelihood ratio test is done by:
 1. $D = -2 \log\{L_{Reduced}(\hat{\beta})\} - (-2 \log\{L_{Full}(\hat{\beta})\})$
 2. Δdf = Number of parameters in the full model - Number of parameters in the reduced model
 3. Reject H_0 : “models are the same” if $D > \chi^2_{\Delta df, 1-\alpha}$, and use the “full” model.
- ▶ When doing a likelihood ratio test with covariance parameters use $\alpha = 0.1$ instead of $\alpha = 0.05$ (the reasons are beyond our scope).

Summary of main points

- ▶ When doing a likelihood ratio test the models **must** be nested.
- ▶ Some people use the REML function in place of the usual likelihood to form likelihood ratio tests and the AIC and BIC criteria. If the test concerns different mean models, this is generally **not** recommended.
- ▶ Use of the AIC and BIC criteria based on the REML objective function to choose among covariance models for the same mean model is often used.